

SPRING BOOT – BOOK API LAB

GET Request, Dynamic Routing & POST Request

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Assignment	Lab 2

1. Introduction

This lab demonstrates a Spring Boot REST API for managing books. The API is tested step-by-step using GET and POST requests and dynamic routing for retrieving a specific book by its ID.

2. API Flow

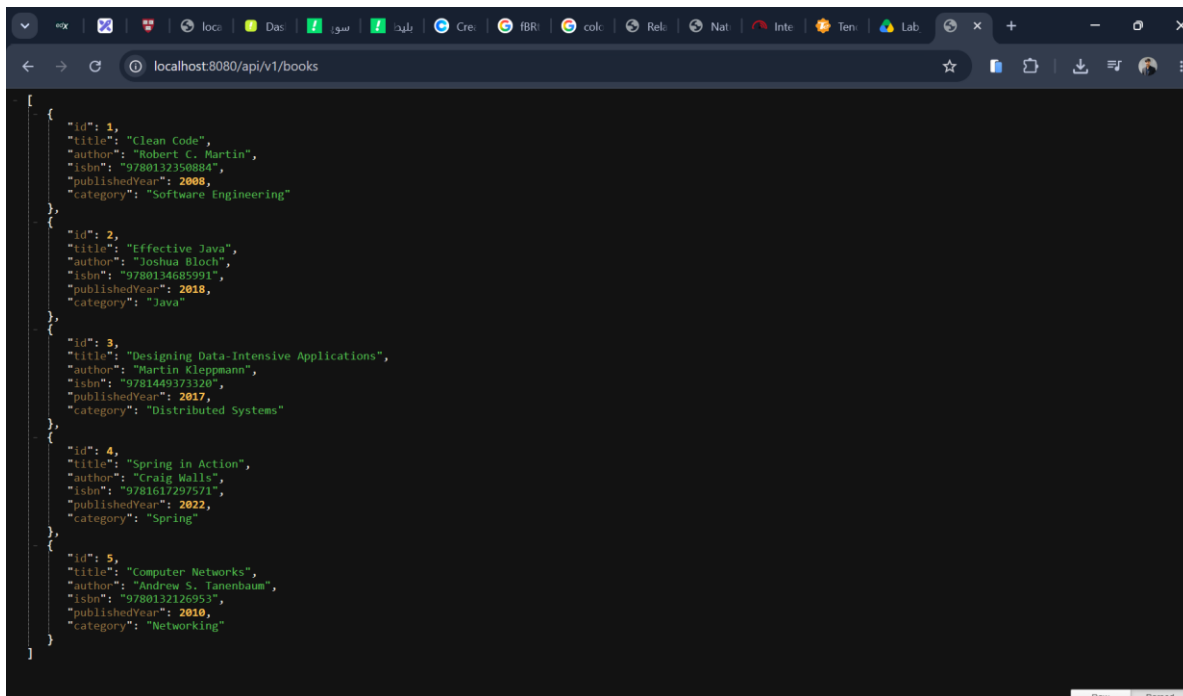
1. GET /api/v1/books – retrieve all books.
2. GET /api/v1/books/{id} – retrieve one specific book using dynamic routing.
3. POST /api/v1/books – add a new book.
4. GET /api/v1/books – retrieve all books again and verify the newly added book.

3. GET Request – Retrieve All Books

The first request uses GET to retrieve all books from the API.

Request: GET /api/v1/books

Expected result: HTTP 200 OK with a JSON list of books.



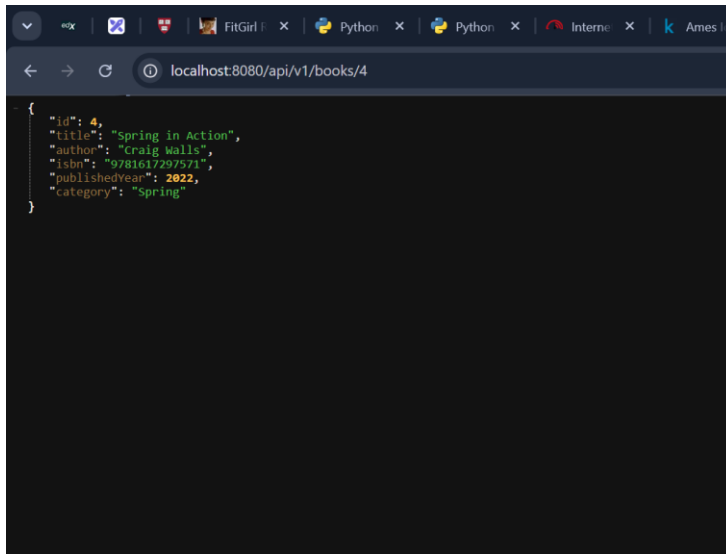
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4. Dynamic Routing – Retrieve One Book

Dynamic routing retrieves one specific book by its ID. The ID is supplied as a path variable in the URL.

Example: GET /api/v1/books/4

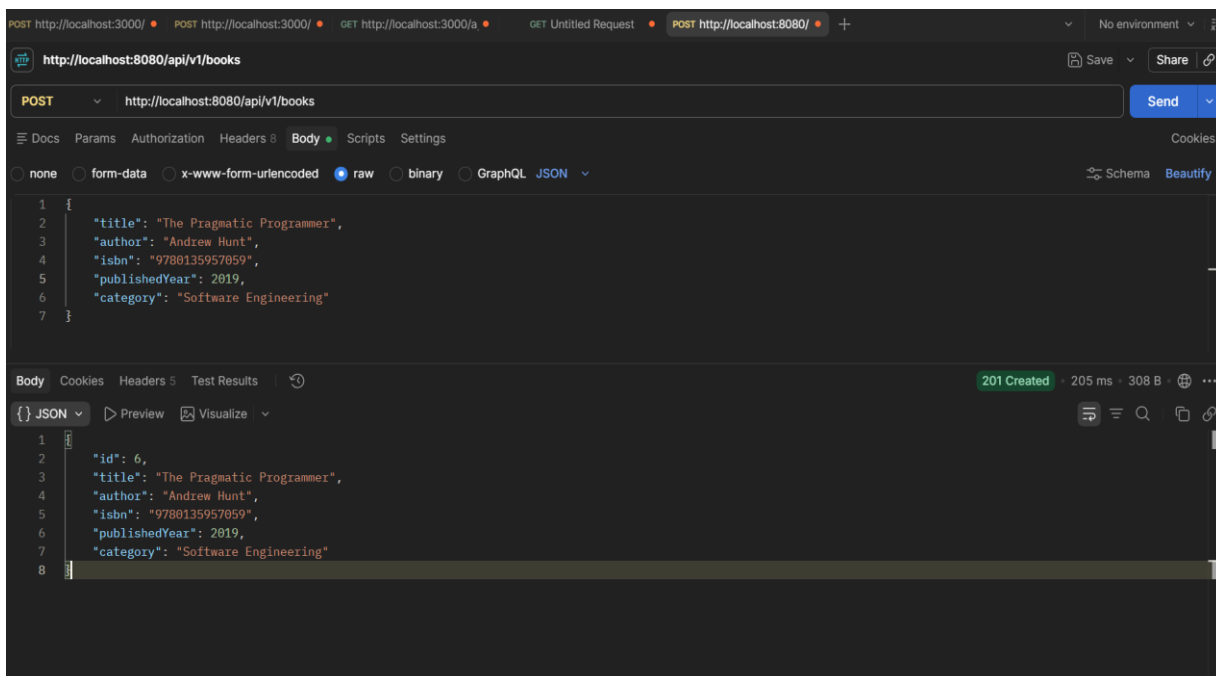
Dynamic route: /api/v1/books/{id}



5. POST Request – Add Another Book

POST is used to send a new book to the server and add it to the collection.

Request: POST /api/v1/books

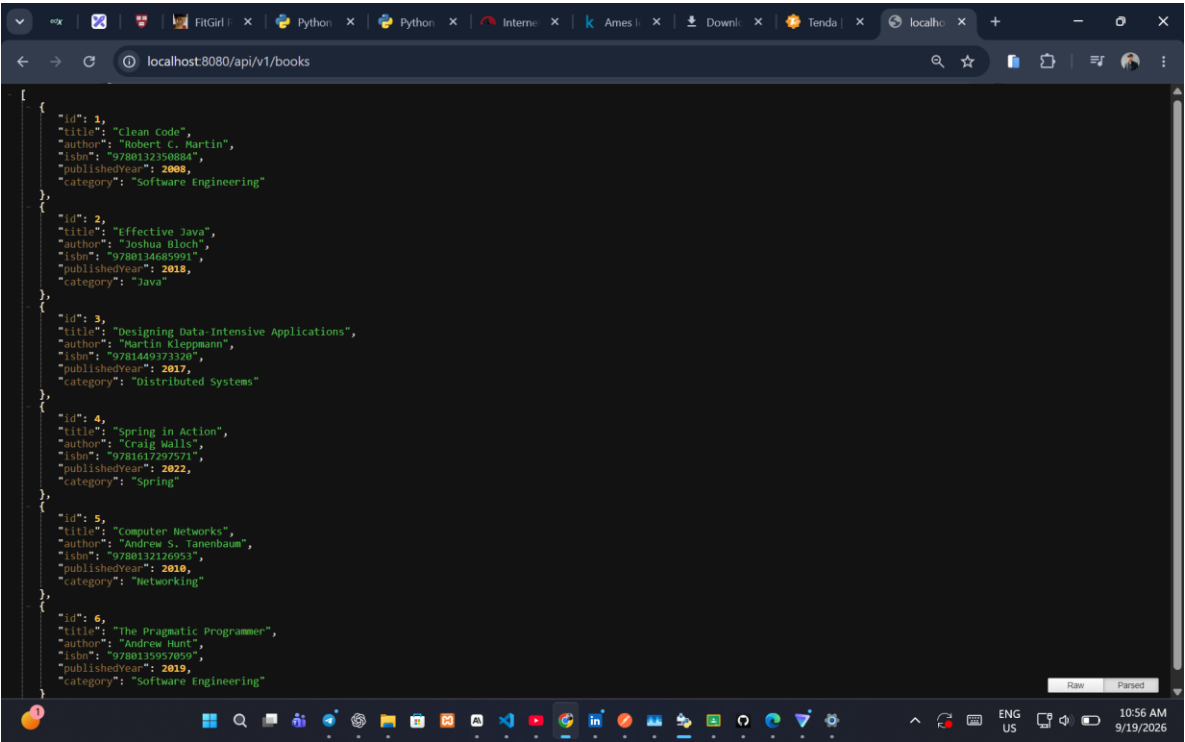


6. Final GET Request – Verify the Added Book

After the POST request, call GET /api/v1/books again. The response should contain the original books plus the newly added book at the end.

Request: GET /api/v1/books

Verification: The new book appears in the returned JSON list.



7. Request Flow Summary

Step	Method	Endpoint	Purpose
1	GET	/api/v1/books	Retrieve all books
2	GET	/api/v1/books/{id}	Retrieve one book using dynamic routing
3	POST	/api/v1/books	Add a new book
4	GET	/api/v1/books	Verify all books including the new book

8. Conclusion

This lab demonstrates a complete Book API workflow: retrieving all books with GET, retrieving one book with dynamic routing, adding a new book with POST, and finally retrieving all books again to verify the new record.